

Vol 4 Chapter 5 — The Hubble Constant via Four Substrate Routes

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Chapter 5 — The Hubble Constant via Four Substrate Routes

Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

The Hubble constant H_0 — the present-day expansion rate of the universe — has been the subject of the famous Hubble tension since 2019. Two independent measurement programs give discrepant values:

- Cosmic-microwave-background (Planck satellite) inference: $H_0 \approx 67.4$ km/s/Mpc
- Distance-ladder (SH0ES collaboration) inference: $H_0 \approx 73.0$ km/s/Mpc

The discrepancy is at the 5σ level and has resisted resolution. Standard physics has produced many proposals (early dark energy, modified gravity, sterile neutrinos, etc.); none has converged on a community consensus.

BST resolves the tension by deriving H_0 from substrate primaries via four independent routes, all consistent at ~ 67 -68 km/s/Mpc — the lower (Planck) value. The framework predicts the higher SH0ES value is a systematic in the distance-ladder calibration rather than evidence of new physics.

5.1 Route A — from baryogenesis η

The substrate's baryon-to-photon ratio η combines with Λ CDM cosmology to give H_0 at ~ 67.5 km/s/Mpc. The substrate-mechanism reading goes through the cosmological Casimir-vacuum identification (Chapter 4) combined with the substrate's natural baryogenesis scale.

5.2 Route B — from $\sqrt{19\Lambda/39}$

A specific substrate-mechanical identity gives $H_0 = \sqrt{19\Lambda/39}$ where Λ is the cosmological constant of Chapter 4 and the integer combination 19/39 uses BST primary structure. Evaluation: $H_0 \approx 67.7$ km/s/Mpc.

5.3 Route C — from full CAMB Toy 677

Elie's Toy 677 ran the substrate-derived Λ CDM parameters through the full CAMB cosmological perturbation code at 0.1% precision. Result: $H_0 \approx 67.4$ km/s/Mpc — exact match to the Planck inference.

5.4 Route D — from $c\sqrt{19\Lambda/39}$

Toy 903 derives $H_0 = c\sqrt{19\Lambda/39}$ as the substrate's natural Hubble expression. Evaluation reproduces the same ~ 67.7 km/s/Mpc value as Route B.

5.5 The Hubble tension resolved

All four substrate routes converge on $H_0 \approx 67$ -68 km/s/Mpc — the low end of the Hubble tension. The substrate predicts the high SH0ES value is a calibration systematic in the distance-ladder rather than a refutation of Λ CDM.

This is one of BST's cleanest dual-camp resolutions. The framework's substrate-derivation is consistent with the Planck inference; the discrepancy with SH0ES is predicted to resolve as the distance-ladder systematics are refined.

5.6 What comes next

Chapter 6 develops CMB structure with the substrate-derived scalar spectral index $n_s = 0.9635$ matching Planck at 0.3σ .

Where to look this up: Toy 677 (CAMB substrate-derived Λ CDM at 0.1%). Toy 903 (substrate-derived Hubble formula). For Hubble tension reviews: Riess 2019 + many subsequent papers; Planck 2018 results VI.